Ps 05

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1 Problem 1

1.1 Task

Let $f : \mathbb{R}^{2n} \to \mathbb{R}$ be defined by $f(x, y) = \|x\|^2 - \|y\|^2$ for $x, y \in \mathbb{R}^n$ and $\|\cdot\|$ the standard norm in $\mathbb{R}^n$.

(a) Determine all local extrema and all saddle points.

(b) Visualize f and ∇f in Julia for $n = 1$.

1.2 Solution

1.2.1 (a)

Let $f : \mathbb{R}^{2n} \to \mathbb{R}, \quad f(x, y) = \|x\|^2 - \|y\|^2$ for $x, y \in \mathbb{R}^n$ and $\|\cdot\|$ the standard norm in $\mathbb{R}^n$.

Hence,

$$\nabla f(x, y) = (2x, -2y) = 0 \iff (x, y) = (0, 0)$$

So $p = (0, 0) \in \mathbb{R}^{2n}$ is the unique critical point; $f(p) = 0$.

Definitions:

1. p is a **local minimum** iff: $\exists \delta > 0 : \forall q \in B_\delta(p), f(q) \geq f(p)$.
2. p is a **local maximum** iff: $\exists \delta > 0 : \forall q \in B_\delta(p), f(q) \leq f(p)$.
3. p is a **saddle** iff: p is a critical point and neither (1) nor (2) holds.

Thus,

Fix $\delta > 0$. Choose $h \in \mathbb{R}^n : 0 < \|h\| < \delta$.

Let $q := (h, 0_{\mathbb{R}^n}) \in \mathbb{R}^{2n}$. Hence, $\|q - p\| = \sqrt{\|h\|^2 + \|0\|^2} = \|h\| < \delta$, so $q \in B_\delta(p)$. $f(q) = \|h\|^2 - \|0\|^2 = \|h\|^2 > 0 = f(p)$. Thus, p is not a local maximum.

Let $q' := (0_{\mathbb{R}^n}, h) \in \mathbb{R}^{2n}$. Hence, $\|q' - p\| = \sqrt{\|h\|^2 + \|0\|^2} = \|h\| < \delta$, so $q' \in B_\delta(p)$. $f(q') = \|0\|^2 - \|h\|^2 = -\|h\|^2 < 0 = f(p)$. Thus, p is not a local minimum.

Hence, p is a saddle. Since it is the unique critical point, f has no local extrema, and $(0, 0)$ is the unique saddle point.

1.2.2 (b)

```
using Plots

# f and f ----------------------------
f(x, y)     = x^2 - y^2
gradf(x, y) = (2x, -2y)

# p1 ----------------------------------
xs = -10:0.05:10
p1 = heatmap(xs, xs, f;
             c = :RdBu,
             aspect_ratio = 1,
             xlims = (-10, 10), ylims = (-10, 10),
             xlabel = "x", ylabel = "y",
             title  = "Level sets and f")

xq   = -10:1:10
pts  = [(x, y) for x in xq, y in xq]
X    = vec([p[1] for p in pts])
Y    = vec([p[2] for p in pts])
UV   = [gradf(p...) for p in pts]
U, V = vec(first.(UV)), vec(last.(UV))
s    = 0.9 * step(xq) / maximum(hypot.(U, V))

quiver!(p1, X, Y; quiver = (s .* U, s .* V), color = :black, linewidth = 1)
scatter!(p1, [0], [0]; color = :black, ms = 5, msw = 0, label = false)

# p2 ----------------------------------
p2 = surface(xs, xs, f;
             alpha  = 0.99,
             color  = :RdBu,
             camera = (30, 30),
             xlabel = "x", ylabel = "y", zlabel = "z",
             title  = "Graph of f")

ts = range(-10, 10, length = 200)
plot3d!(p2, ts, zeros(length(ts)),  ts.^2;  color = :red,  lw = 3, label = "f(t,0) = t²")
plot3d!(p2, zeros(length(ts)), ts, -ts.^2;  color = :blue, lw = 3, label = "f(0,t) = -t²")
scatter!(p2, [0], [0], [0];
         color = :black, markerstrokecolor = :white, markerstrokewidth = 1.5,
         ms = 6, label = "saddle")

# plot -------------------------------
plot(p1, p2; layout = (1, 2), size = (1200, 600))
savefig("./assets/p_01_b.png")
```

2 Problem 2

2.1 Task

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. We consider the loss function

$$L : \mathbb{R}^n \to \mathbb{R}, \quad x \mapsto \|Ax - b\|^2$$

(a) Compute the Fréchet derivative of L.

(b) Show that $x \in \mathbb{R}^n$ is a global minimum if and only if

$$A^* A x = A^* b$$

(c) Implement the (affine) local approximation of L in Julia.

2.2 Solution

2.2.1 (a)

For $x, h \in \mathbb{R}^n$:

$$L(x+h) = \|A(x+h) - b\|^2 = \|(Ax - b) + Ah\|^2$$

Expanding via $\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$:

$$L(x+h) = \|Ax - b\|^2 + 2\langle Ax - b, Ah\rangle + \|Ah\|^2$$

Using the disjoint identity $\langle Ax - b, Ah\rangle = \langle A^*(Ax - b), h\rangle$:

$$L(x+h) - L(x) = 2\langle A^*(Ax - b), h\rangle + \|Ah\|^2$$

The map $h \to 2\langle A^*(Ax - b)$ is linear and bounded. The remainder satisfies:

$$\frac{\|Ah\|^2}{\|h\|} \leq \|A\|_{\text{op}}^2 \|h\| \xrightarrow{h \to 0} 0$$

So $\|Ah\|^2 = o(\|h\|)$ by Definition of Frechet derivative,

$$\boxed{DL(x)[h] = 2\langle A^*(Ax - b),\, h\rangle, \qquad \nabla L(x) = 2A^*(Ax - b).}$$

2.2.2 (b)

Setting $y = x + h$ in the expansion of (a):

$$L(y) - L(x) = 2\langle A^*(Ax - b), y - x\rangle + \|A(y - x)\|^2 \qquad (\star)$$

($\Leftarrow$) Assume $A^*Ax = A^*b$, i.e. $A^*(Ax - b) = 0$. Then $(\star)$ gives

$$L(y) - L(x) = \|A(y - x)\|^2 \geq 0 \quad \text{for all } y \in \mathbb{R}^n,$$

so x is a global minimum.

($\Rightarrow$) Assume x is a global minimum. Since L is Fréchet differentiable, $DL(x) = 0$, hence

$$\langle A^*(Ax - b), h\rangle = 0 \quad \text{for all } h \in \mathbb{R}^n.$$

Taking $h = A^*(Ax - b)$ gives $\|A^*(Ax - b)\|^2 = 0$, so $A^*(Ax - b) = 0$, i.e. $A^*Ax = A^*b$. ■

2.2.3 (c)

The first-order Taylor approximation of L at x_0 is

$$\widehat{L}_{x_0}(x) = L(x_0) + DL(x_0)[x - x_0] = \|Ax_0 - b\|^2 + 2\langle A^*(Ax_0 - b), x - x_0\rangle.$$

```
using LinearAlgebra, Random

let
    function affine_approx(A, b, x0, x)
        r0 = A*x0 - b
        g  = 2 * (A' * r0)
        return dot(r0, r0) + dot(g, x - x0)
    end

    Random.seed!(42)
    A  = randn(20, 3)
    b  = randn(20)
    x0 = randn(3)

    L(x) = sum(abs2, A*x - b)

    println(L(x0))
    println(affine_approx(A, b, x0, x0))
```

```
    dx = 1e-3 * randn(3)
    println(L(x0 + dx) - affine_approx(A, b, x0, x0 + dx))
    println(sum(abs2, A*dx))
end
```

3 Problem 3

3.1 Task

Consider the functions $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}, \quad g(u, v) = u - v$$

Compute $(g \circ f)'(x, y)$.

3.2 Solution

$$\begin{aligned} f'(x, y) &= \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix} \\ g'(u, v) &= (1, -1) \\ (g \circ f)'(x, y) &= (1, -1) \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix} \\ &= (2x, -2y) \end{aligned}$$

4 Problem 4

4.1 Task

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$f(x, y) = \begin{pmatrix} xy \\ x^2 + y^3 \end{pmatrix}$$

(a) Compute the Jacobi matrix $J_f(x, y)$.

(b) Evaluate $J_f(1, 1)$.

(c) Determine the (affine) linear approximation of f near the point $(1, 1)$.

(d) Use this approximation to estimate $f(1.01, 0.98)$.

4.2 Solution

4.2.1 (a)

Define $f_1(x, y) = xy, f_2(x, y) = x^2 + y^3$. Hence,

$$\begin{aligned} f(x, y) &= \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} \\ \implies J_f(x, y) &= \begin{pmatrix} \partial_x f_1 & \partial_y f_1 \\ \partial_x f_2 & \partial_y f_2 \end{pmatrix} \\ &= \begin{pmatrix} y & x \\ 2x & 3y^2 \end{pmatrix} \end{aligned}$$

4.2.2 (b)

$$\begin{aligned} J_f(x, y) &= \begin{pmatrix} y & x \\ 2x & 3y^2 \end{pmatrix} \\ J_f(1, 1) &= \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \end{aligned}$$

4.2.3 (c)

Linear approximation of $f : \mathbb{R}^n \to \mathbb{R}^m$ at x_0 is

$$L_f(x) = f(x_0) + J_f(x_0)(x - x_0)$$

Define $x_0 = (1, 1)$. Thus,

$$L_f(x, y) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x - 1 \\ y - 1 \end{pmatrix}$$

4.2.4 (d)

$$
\begin{aligned}
L_f(x,y) &= \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} x-1+y-1 \\ 2x-2+3y-3 \end{pmatrix} \\
&= \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} x+y-2 \\ 2x+3y-5 \end{pmatrix} \\
&= \begin{pmatrix} x+y-1 \\ 2x+3y-3 \end{pmatrix} \\
L_f(1.01, 0.98) &= \begin{pmatrix} 1.01+0.98-1 \\ 2(1.01)+3(0.98)-3 \end{pmatrix} \\
&= \begin{pmatrix} 0.99 \\ 1.96 \end{pmatrix}
\end{aligned}
$$